Power Input & Protection

File: power_in.kicad_sch

Charger, Battery & Rails

File: power_rails.kicad_sch

MCU, USB, Flash & Debug

File: mcu.kicad_sch

LTE-M / NB-IoT Modem

File: cellular.kicad_sch

GNSS L1/L5 + NavIC

File: gnss.kicad_sch

Sensors & Vehicle I/O

File: io.kicad_sch

```
CRL-TRK-2W-01  AIS-140 two-wheeler tracker  -  60 x 45 mm, 4-layer 1.6 mm (JLC04161H-7628)

VEHICLE 9-36 V -> F1/D1 -> LM74700 ideal diode -> pi filter -> LM5164 buck 5.0 V -+-> BQ24074 power path -> VSYS 3.0-4.4 V
USB-C VBUS 5 V ------------------------------------------------------------------+    |-> TPS22917 -> SIM7080G VBAT
18650 1S pack (PCM + NTC) <-> BQ24074 BAT  (LC709203F fuel gauge, VBAT/2 ADC)      |-> TLV75733 -> +3V3 (always on)
                                                                                  -> TPS22917 -> +3V3_GNSS

ESP32-S3-WROOM-1 <-UART(1.8 V via TXS0108E)-> SIM7080G LTE-M/NB-IoT  -> U.FL -> flex LTE antenna
                 <-UART---------------------> NEO-F10N L1/L5 + NavIC  -> U.FL -> active L1/L5 patch
                 <-I2C--> IIM-42652 IMU, LC709203F gauge, RV-3028 RTC     <-SPI-> W25Q128 log flash
                 <-GPIO-> IGN, PANIC, DIN1, AIN1, DOUT1 (relay), LEDs, Tag-Connect JTAG, USB-C

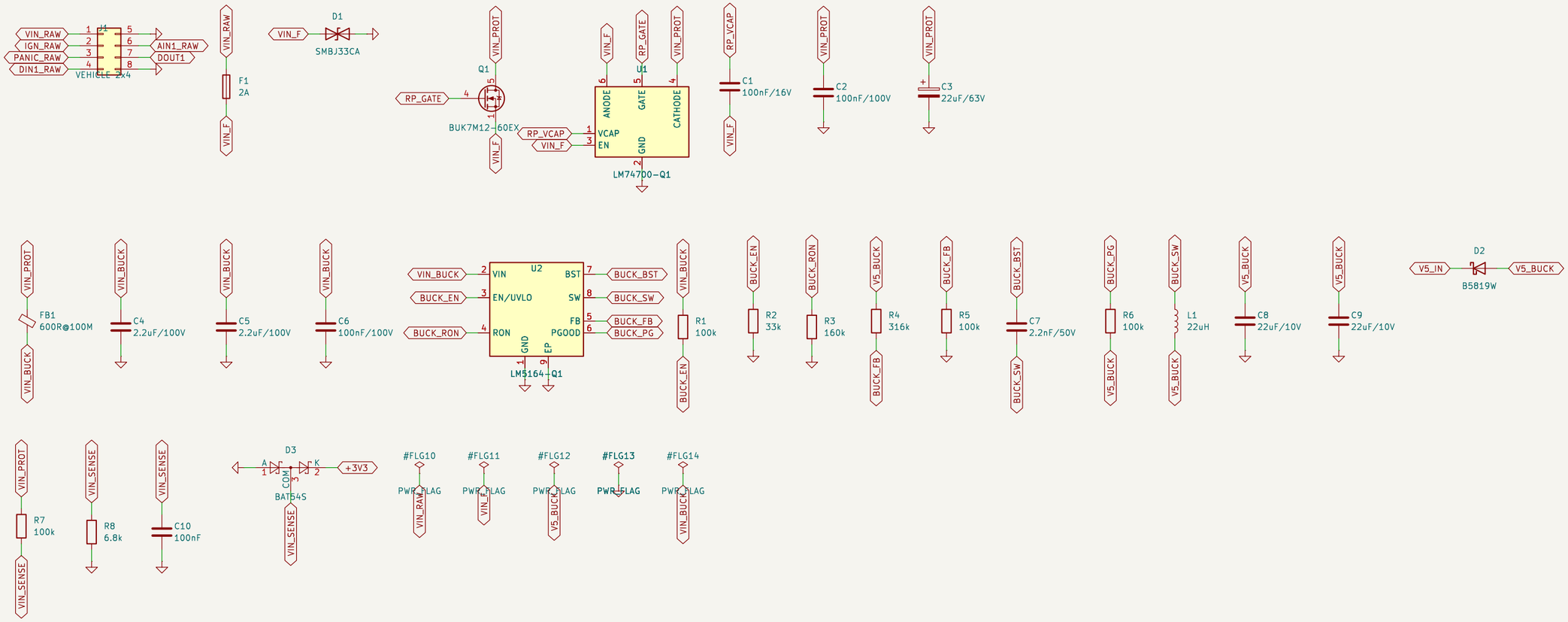
Sleep budget (ignition off, modem PSM, GNSS backup): ~110 uA from the pack - see 04_docs/POWER_TREE.md
```

Top level: block diagram
Generated from 02_python/design.py - edit the source, then regenerate
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Sheet: /
File: tracker.kicad_sch

Title: CRL-TRK-2W-01 - Top level

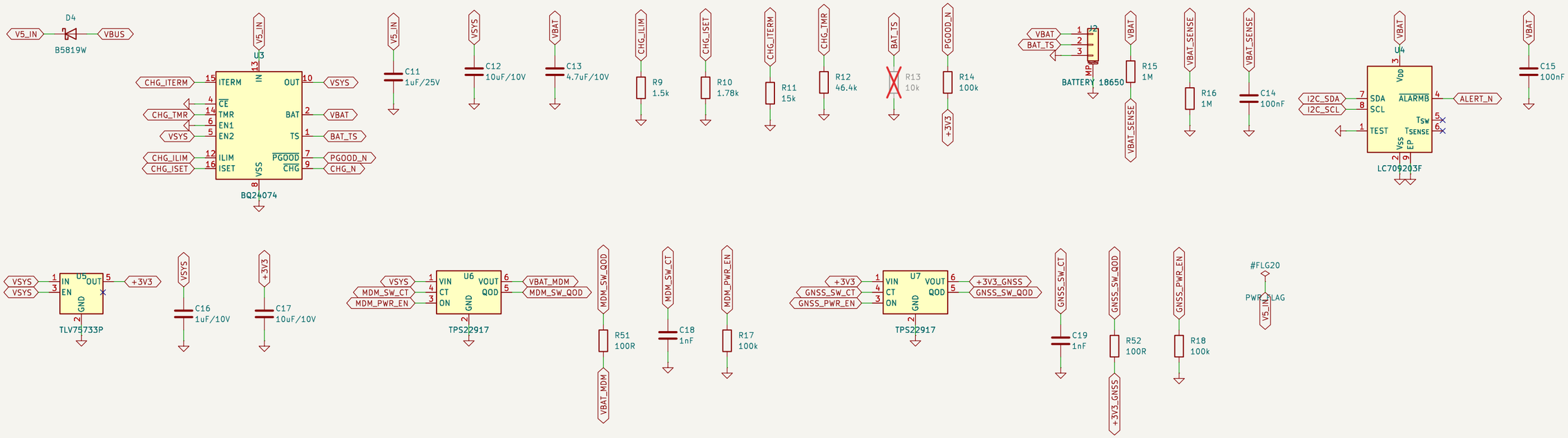
Size: A3	Date: 2026-10-17	**Rev: A0**
KiCad E.D.A. 9.0.9		Id: 1/7



DESIGN NOTES
Input 9–36 V (12 V nominal, survives 24 V jump start, −33 V reverse, ISO 7637-2 pulses via D1).
F1 + D1 at the connector, then LM74700-Q1 ideal diode (Q1): <15 mV drop, blocks reverse battery.
FB1 + C4/C5 pi filter keeps buck ripple off the harness (CISPR 25).
LM5164-Q1 100 V COT buck: 5.0 V / 1 A, 10 uA IQ (key for parked-vehicle battery drain).
Layout: buck loop (C4–U2–L1–C8) <10 mm, SW node small, far from GNSS (see PCB notes).

PART NOTES (VERIFY = check against vendor datasheet before release)
J1: 1 +12V(BAT) 2 IGN 3 PANIC(SOS, to GND) 4 DIN1 5 GND 6 AIN1 7 DOUT1(relay low side) 8 GND
F1: VERIFY MPN suffix
U1: EN tied to ANODE (always on)
C1: VCAP to ANODE
FB1: VERIFY MPN (BLM31 automotive)
R1: UVLO start ~6.0 V with R2 (EN 1.5 V) VERIFY
R3: RON sets COT fsw ~300 kHz – VERIFY with LM5164 eq.
R4: VOUT = 1.2 V x (1 + R4/R5) = 5.0 V
R8: VIN 0–50 V –> 0–3.2 V (ADC1_CH0)

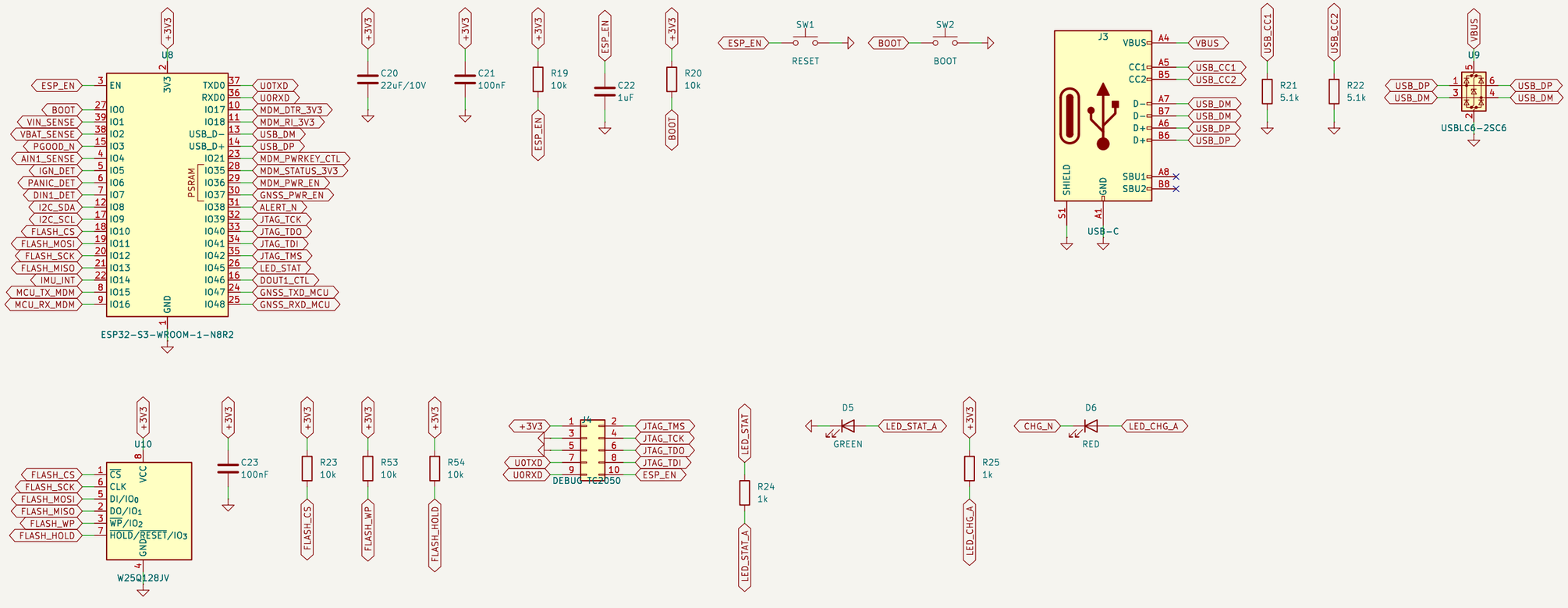
Sheet 2: Power Input & Protection
Generated from 02_python/design.py – edit the source, then regenerate
CRL / BEL – RFM Lab
Sheet: /Power Input & Protection/
File: power_in.kicad_sch
Title: CRL-TRK-2W-01 – Power Input & Protection
Size: A3 | Date: 2026-10-17 | **Rev: A0**
KiCad E.D.A. 9.0.9 | Id: 2/7



DESIGN NOTES
BQ24074: power path. OUT (VSYS) = 4.4 V on external power, = VBAT on battery.
Charging is blocked outside 0–45 C by the pack NTC on TS (parked-in-sun safety).
VSYS feeds the modem directly (SIM7080G VBAT 2.7–4.8 V, 0.5 A peaks) via U6 load switch,
which lets firmware hard-reset a hung modem. U5 3.3 V LDO is always on (deep sleep rail).
U7 switches GNSS VCC; V_BCKP stays on +3V3 so ephemeris is kept (hot start < 2 s).

PART NOTES (VERIFY = check against vendor datasheet before release)
U3: EN2=1/EN1=0 -> ILIM resistor mode; CE low = charge enabled
R9: IIN_MAX = 1550/R = 1.03 A VERIFY K_ILIM
R10: ICHG = 890/R = 500 mA VERIFY K_ISET
R11: termination ~10 % ICHG VERIFY
R12: safety timer VERIFY (open = default timer)
R13: fit only when the pack has no NTC
R16: VBAT/2 -> ADC1_CH1, 2.1 uA
U4: Temperature via I2C (host writes IMU temp); TSW/TSENSE unused VERIFY
U6: QOD via R to VOUT = quick output discharge

Sheet 3: Charger, Battery & Rails
Generated from 02_python/design.py — edit the source, then regenerate
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Sheet: /Charger, Battery & Rails/
File: power_rails.kicad_sch
Title: CRL-TRK-2W-01 — Charger, Battery & Rails
Size: A3 | Date: 2026-10-17 | **Rev: A0**
KiCad E.D.A. 9.0.9 | Id: 3/7



DESIGN NOTES
ESP32-S3: native USB (IO19/20) gives flashing + JTAG + CDC log without extra parts.
TC2050 Tag-Connect for production JTAG/UART; BOOT/RESET buttons for bench use.
Strapping: IO0 boot (pull-up), IO45 LED to GND (VDD_SPI=3.3 V ok), IO46 held low by R50.
RTC-capable wake pins: IO5 IGN, IO6 PANIC, IO14 IMU INT, IO18 modem RI.

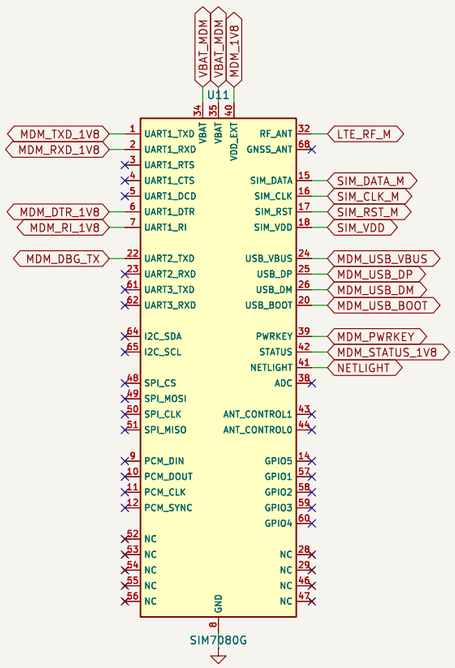
PART NOTES (VERIFY = check against vendor datasheet before release)
U8: Quad PSRAM keeps IO35-37 free

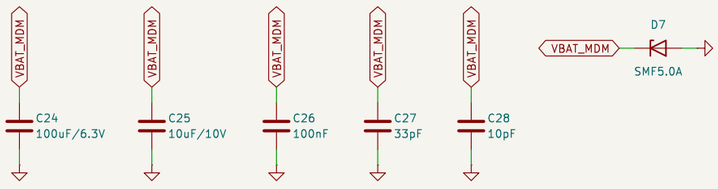
Sheet 4: MCU, USB, Flash & Debug
Generated from 02_python/design.py - edit the source, then regenerate
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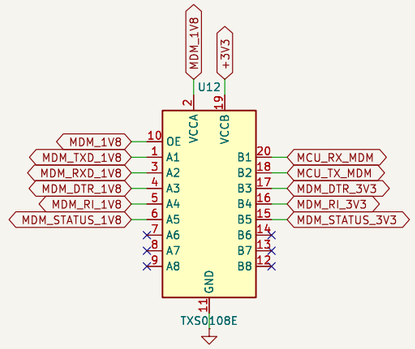
Sheet: /MCU, USB, Flash & Debug/
File: mcu.kicad_sch

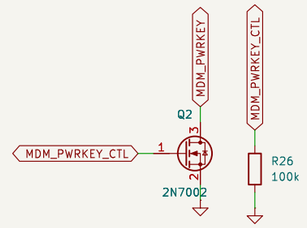
Title: CRL-TRK-2W-01 - MCU, USB, Flash & Debug

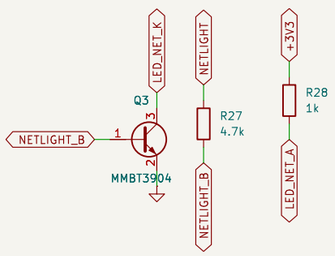
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KiCad E.D.A. 9.0.9		Id: 4/7

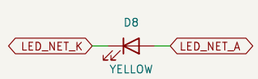


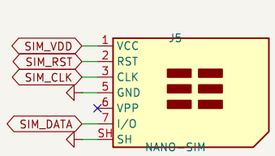


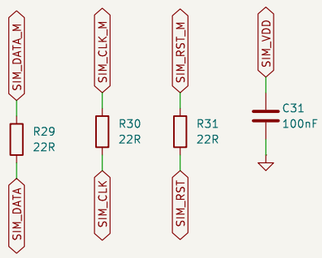


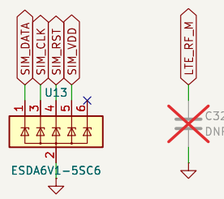


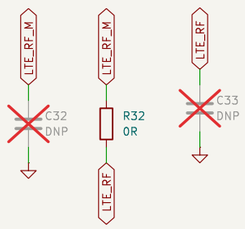


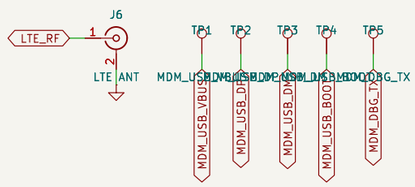










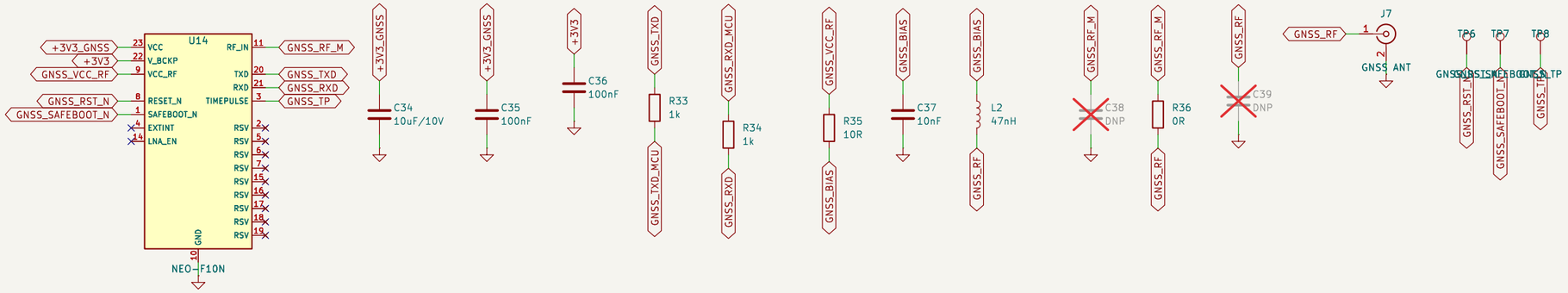


DESIGN NOTES
SIM7080G: Cat-M1/NB-IoT, PSM 3 uA. 1.8 V UART via TXS0108E (OE = VDD_EXT).
PWRKEY driven open-drain by Q2. Never cut VBAT while the module runs: AT+CPOWD first,
U6 only for watchdog recovery. RF_ANT -> 50 R CPWG -> pi pad -> U.FL (flex antenna).
SIM: 22 R series + ESD array next to the holder; keep SIM traces away from RF and VBAT.
TP1-TP5: modem USB (firmware upgrade, logs) - bring to bed-of-nails fixture.

PART NOTES (VERIFY = check against vendor datasheet before release)
U11: Internal GNSS unused (no NavIC): NEO-F10N used instead
J5: AIS-140 production: replace by MFF2 eSIM on the same SIM bus (rev B footprint)

Sheet 5: LTE-M / NB-IoT Modem
Generated from 02_python/design.py - edit the source, then regenerate
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Sheet: /LTE-M / NB-IoT Modem/
File: cellular.kicad_sch
Title: CRL-TRK-2W-01 - LTE-M / NB-IoT Modem

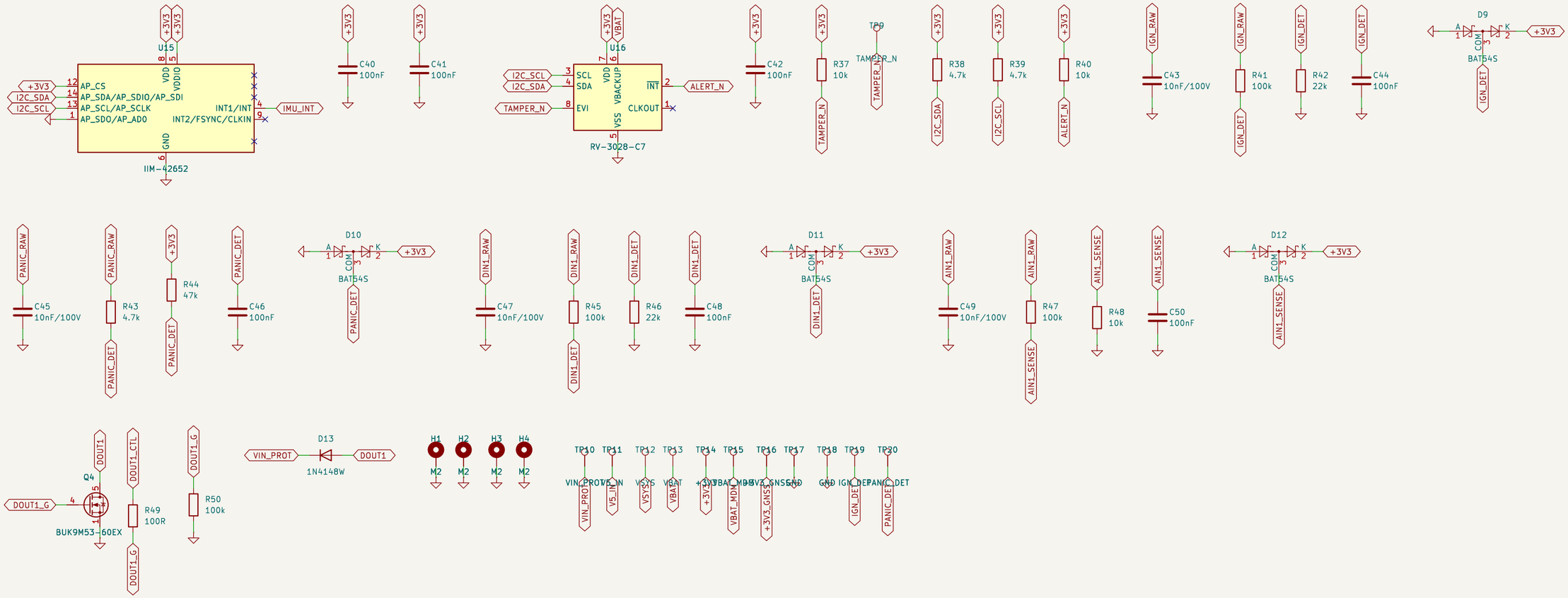
Size: A3	Date: 2026-10-17	**Rev: A0**
KiCad E.D.A. 9.0.9		Id: 5/7



DESIGN NOTES
NEO-F10N dual-band L1/L5: NavIC SPS-L5 required by AIS-140. RF_IN has an internal DC block.
Active antenna bias from VCC_RF through R35/L2 (u-blox IM fig.6); DNP R35+L2 for passive antennas.
RF trace: 50 R CPWG on L1 over solid L2 ground, via fence every 1.5 mm, no other nets nearby.
UART 1 k series resistors stop back-powering while GNSS VCC is switched off.

PART NOTES (VERIFY = check against vendor datasheet before release)
C36: V_BCKP: always-on for hot start
R33: limits back-power while GNSS VCC is off
R35: u-blox R17 10 R short-circuit limiter
L2: DNP L2+R35 for a passive antenna

Sheet 6: GNSS L1/L5 + NavIC
Generated from 02_python/design.py — edit the source, then regenerate
CRL / BEL — RFM Lab
Sheet: /GNSS L1/L5 + NavIC/
File: gnss.kicad_sch
Title: CRL-TRK-2W-01 — GNSS L1/L5 + NavIC
Size: A3 | Date: 2026-10-17 | **Rev: A0**
KiCad E.D.A. 9.0.9 | Id: 6/7



DESIGN NOTES
Inputs survive 0–60 V and reverse battery: 1206 series resistors (200 V), BAT54S clamps, RC filter.
PANIC: SOS switch to chassis GND (AIS-140 emergency button). IGN/DIN1: 12/24 V active high.
DOUT1: low-side immobiliser relay driver, flyback D13 to VIN_PROT.
IMU wake-on-motion -> IO14 (theft alert), crash = >6 g peak + tilt > 60 deg (firmware).
RV-3028 EVI pin time-stamps lid opening (TAMPER_N) even with the MCU asleep.

PART NOTES (VERIFY = check against vendor datasheet before release)
U15: Wake-on-motion, tilt/fall + crash (+-16 g) detection
R42: 14.4 V -> 2.6 V; IGN > ~7 V reads high
R48: 0-30 V -> 0-2.73 V (ADC1_CH3)
R50: holds IO46 strap low at boot

Sheet 7: Sensors & Vehicle I/O
Generated from 02_python/design.py – edit the source, then regenerate
CRL / BEL – RFM Lab
Sheet: /Sensors & Vehicle I/O/
File: io.kicad_sch

Title: CRL-TRK-2W-01 – Sensors & Vehicle I/O

Size: A3	Date: 2026-10-17	**Rev: A0**
KiCad E.D.A. 9.0.9		Id: 7/7